

# Machine Learning Fundamentals

Machine learning is a branch of artificial intelligence that enables systems to learn patterns from data and make predictions or decisions.

Supervised learning uses labeled examples. Common supervised learning tasks include classification and regression.

Classification predicts a discrete category, such as spam or not spam.

Regression predicts a continuous numerical value, such as house price.

Unsupervised learning works with data without target labels. Clustering is a common unsupervised learning technique.

Overfitting happens when a model learns the training data too closely and performs poorly on unseen data.

A train-test split separates data into training and evaluation sets so that generalization can be measured.